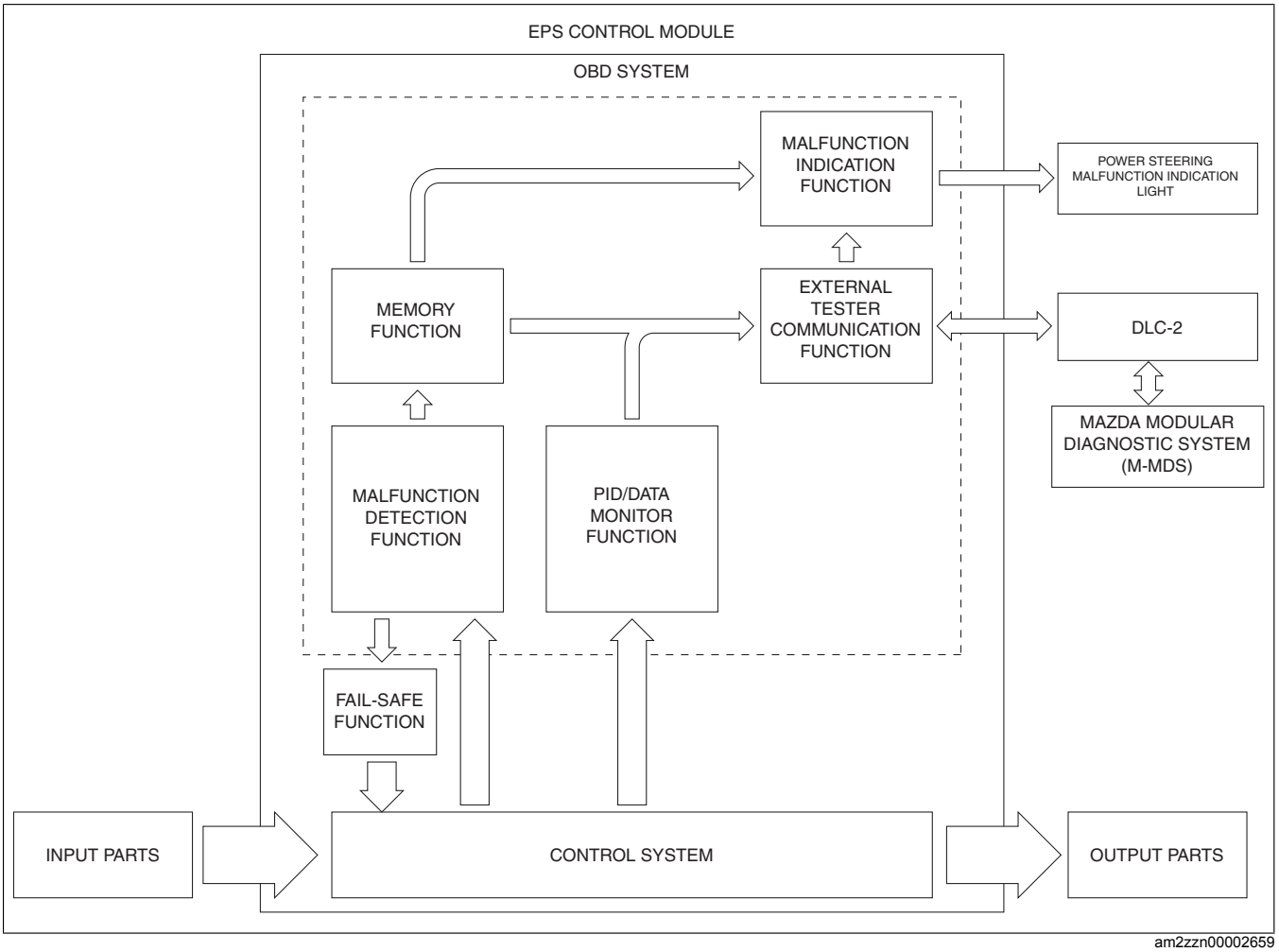
ON-BOARD DIAGNOSTIC SYSTEM [EPS CONTROL MODULE]

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Outline

- The on-board diagnostic system consists of the following functions:
  - — A malfunction detection function that detects abnormalities in input/output signals when the ignition switch is at the ON position.
  - — A PID/data monitor function that reads out specified input/output signals.
- The data link connector 2 (DLC-2), which groups together all the connectors used for malfunction diagnosis into a single location, has been adopted, thereby improving serviceability. Diagnosis is performed by connecting the Mazda Modular Diagnostic System (M-MDS) to the DLC-2.
- In addition to DTC read-out, the Mazda Modular Diagnostic System (M-MDS) has following functions:
  - — PID/data monitor access
  - — DTC clear using the display screen of the diagnostic tester

Block diagram



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Function

Malfunction detection function

- The malfunction detection function detects malfunctions in the input/output signal system of the EPS control module when the ignition switched ON or driving the vehicle.

Malfunction display function

- When the malfunction detection function detects a malfunction, the power steering malfunction indicator light illuminates or flashes to alert the driver.
- Using the external tester communication function, DTCs can be output to the DLC-2 via the CAN line. At the same time, malfunction detection results are sent to the memory function and fail-safe function.

Memory function

- The memory function stores DTCs of malfunctions in input/output signal systems. With this function, once a DTC is stored it is not cleared after the ignition switch has been switched OFF, even if the malfunctioning signal system has returned to normal.

- Since the EPS control module has a built-in non-volatile memory, DTCs are not cleared even if the battery is removed. Therefore, it is necessary to clear the memory after performing repairs. Refer to the Workshop Manual for the DTC clearing procedure.

DTC table

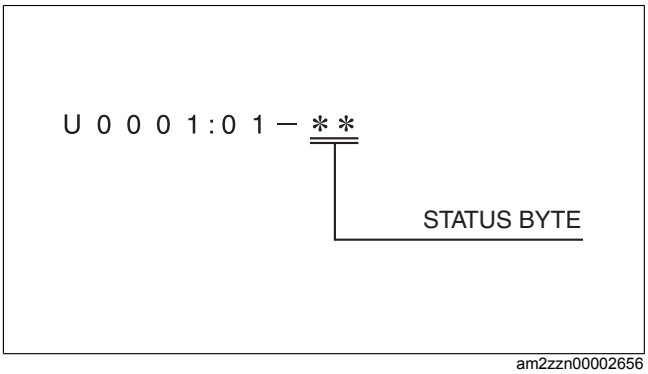
×: Applicable
—: Not applicable

DTC	Power steering malfunction indicator light	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
C1B00:2F	Not illuminate	Steering angle sensor	×	—	C	×
C1B00:54	Illuminates/flashes	Steering angle sensor	×	—	C	×
C1B00:61	Illuminates	Steering angle sensor	×	—	C	×
C1B00:62	Illuminates	Steering angle sensor	×	—	C	×
C1B00:95	Illuminates	Steering angle sensor	×	—	C	×
C200B:2F	Illuminates	Torque sensor	×	—	C	×
C200B:61	Illuminates	Torque sensor	×	—	C	×
C200B:62	Illuminates	Torque sensor	×	—	C	×
C200C:2F	Illuminates	Torque sensor	×	—	C	×
C200D:49	Illuminates	EPS motor	×	—	C	×
U0001:88	Not illuminate	CAN communication: module communication error	×	—	C	×
U0100:00	Illuminates	CAN communication: communication error to PCM	×	—	C	×
U0121:00	Not illuminate	CAN communication: communication error to DSC HU/CM	×	—	C	×
U0155:00	Not illuminate	CAN communication: communication error to instrument cluster	×	—	C	×
U0214:00	Not illuminate	CAN communication: communication error to start stop unit	×	—	C	×
U0338:00	Not illuminate	CAN communication: signal error from start stop unit	×	—	C	×
U0401:00	Illuminates	CAN communication: signal error from PCM (engine speed)	×	—	C	×
	Illuminates	CAN communication: signal error from PCM (vehicle speed)	×	—	C	×
	Not illuminate	CAN communication: signal error from PCM (i-stop condition)	×	—	C	×
U0415:00	Not illuminate	CAN communication: signal error from DSC HU/CM	×	—	C	×
U0515:00	Not illuminate	CAN communication: signal error from start stop unit	×	—	C	×
U2011:49	Illuminates	EPS motor	×	—	C	×
U2011:61	Illuminates	EPS motor	×	—	C	×
U2300:54	Not illuminate	EPS configuration	×	—	C	×
U2300:55	Illuminates	EPS configuration	×	—	C	×
U2300:56	Not illuminate	EPS configuration	×	—	C	×
U3000:05	Illuminates	EPS control module	×	—	C	×
U3000:41	Illuminates	EPS control module	×	—	C	×
U3000:43	Not illuminate	EPS control module	×	—	C	×
U3000:46	Illuminates	EPS control module	×	—	C	×
U3000:49	Illuminates	EPS control module	×	—	C	×
U3000:61	Illuminates	EPS control module	×	—	C	×
U3000:96	Illuminates	EPS control module	×	—	C	×
U3003:16	Illuminates	Battery power supply	×	—	C	×
U3003:17	Illuminates	Battery power supply	×	—	C	×
U3004:49	Illuminates	EPS control module	×	—	C	×

*1 : C: CMDTC self test

Status byte for DTC

- The status byte is the two-digit code (two digits after hyphen (-)) after the seven-digit DTC.
- The status byte is a code which indicates the pending code, current/past malfunction status, or warning illumination status.
- The status byte can be read by performing a CMDTC self test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the M-MDS when reading the DTC.



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Fail-safe function

- When the malfunction detection function determines a malfunction, the power steering malfunction indicator light illuminates or flashes to alert the driver. At this time, the fail-safe function disables system control or gradually reduces the assist power.

Fail-safe function table

DTC	EPS motor current control
C1B00:2F	Control enabled
C1B00:54	Control enabled
C1B00:61	Control disabled
C1B00:62	Control enabled
C1B00:95	Control enabled
C200B:2F	Control disabled
C200B:61	Control disabled
C200B:62	Control disabled
C200C:2F	Control disabled
C200D:49	Control disabled/Control is maintained in the backup control
U0001:88	Control enabled
U0100:00	Control is maintained by gradually changing the controlled vehicle speed and setting it to specified speed
U0121:00	Control enabled
U0155:00	Control enabled
U0214:00	Control enabled
U0338:00	Control enabled
U0401:00	• Engine speed — Control enabled • Vehicle speed — Control is maintained by gradually changing the controlled vehicle speed and setting it to specified speed • i-stop condition — Control enabled
U0415:00	Control enabled
U0515:00	Control enabled
U2011:49	Control disabled/Control is maintained in the backup control
U2011:61	Control disabled
U2300:54	Control enabled
U2300:55	Control enabled
U2300:56	Control enabled
U3000:05	Control disabled
U3000:41	Control disabled
U3000:43	Control enabled
U3000:46	Control disabled
U3000:49	Control disabled
U3000:61	Control disabled
U3000:96	Control disabled
U3003:16	Control is maintained by gradually decreasing the motor control current. However, control is inhibited if the power supply voltage is the specified value of less
U3003:17	Control disabled

DTC	EPS motor current control
U3004:49	Control disabled

Snapshot data

- The snapshot data stores the currently detected DTC data.

—: Not applicable

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
AAT	°C, °F	Ambient air temperature	—	—
APP_STATUS	Accelerator Pedal Off/ Under 20%/ Over 20%/ FAIL	Accelerator pedal position	—	—
CEN_TRQ_S	Nm	Center value of torque sensor	—	—
CFG_STATUS	Config Complete/ Not Configured/ Config Error	Configuration status	—	—
ECT_STATUS	Under 0 degrees C/ 0 - Under 80 degrees C/ Over 80 degrees C/ FAIL	Engine coolant temperature status	—	—
ECU_IN_TMP	°C, °F	ECU internal temperature	—	ECU_IN_TMP
ENG_RPM	RPM	Engine speed	—	ENG_RPM
IC_VPWR	V	Instrument cluster power supply	• The EPS control module constantly receives the power supply voltage value of the instrument cluster sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the power supply voltage of the instrument cluster when the DTC was detected, and it is displayed in the M-MDS.	—
IG-ON_TIMER	hh:mm:ss*1	Elapsed time since ignition was switched ON **Note** • The instrument cluster records the elapsed time since the ignition was switched ON.	• The EPS control module constantly receives the elapsed time since the ignition was switched ON sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the elapsed time since the ignition was switched ON when the DTC was detected, and it is displayed in the M-MDS.	—
MT_CURRENT	A	EPS motor control current	—	MT_CURRENT

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
OH_CR_C	No/Yes	Current overheat protection control (complete)	—	OH_CR_C
OH_CR_M	No/Yes	Current overheat protection control (middle)	—	OH_CR_M
OH_HIST_C	No/Yes	History of overheat protection control (complete)	—	OH_HIST_C
OH_HIST_M	No/Yes	History of overheat protection control (middle)	—	OH_HIST_M
OH_IG_CNT_C	—	The number of times the ignition is switched ON during the period from when overheat protection control (complete) is finished to the time the next overheat protection control (complete) is started • Displays a maximum number of 255 times in which the ignition is switched ON until the first overheat protection control (complete) is started. • During overheat protection control (complete), the number of times the ignition is switched ON is not counted, and 0 is displayed.	—	OH_IG_CNT_C
OH_IG_CNT_M	—	The number of times the ignition is switched ON during the period from when overheat protection control (middle) is finished to the time the next overheat protection control (middle) is started • Displays a maximum number of 255 times in which the ignition is switched ON until the first overheat protection control (middle) is started. • During overheat protection control (middle), the number of times the ignition is switched ON is not counted, and 0 is displayed.	—	OH_IG_CNT_M

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
PWR_MODE_KEY	Key Out/Key Recently Out/Key Approved (Position 0)/Post Accessory (Position 0)/ Accessory (Position 1)/ Post Ignition (Position 1)/Ignition On (Position 2)/Running (Position 2)/ Running - Starting In Progress (Position 2)/ Crank (Position 3)	• Key Out: Ignition switched off • Key Recently Out (Position 0): Elapsed time within 3 s since ignition was switched off • Accessory (Position 1): Ignition is switched to ACC • Post Ignition (Position 2): Elapsed time within 3 s since ignition was switched ON • Ignition On (Position 2): Ignition switched ON (engine off) • Running (Position 2): Ignition switched ON (engine on) • Running - Starting: Cranking condition	• The EPS control module constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the ignition switch status when the DTC was detected, and it is displayed in the M-MDS.	—
RPM_STATUS	Engine Stop/ Under 1500 rpm/ Over 1500 rpm/ FAIL	Engine RPM status	• The EPS control module constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the ignition switch status when the DTC was detected, and it is displayed in the M-MDS.	—
SHIFT_STATUS	P/N D/ R/ FAIL	Shift position status	• The EPS control module constantly receives the selector lever position sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the selector lever position when the DTC was detected, and it is displayed in the M-MDS.	—
STR_ANG	°	Steering wheel angle	—	STR_ANG
STR_ANG_EST	°	Steering wheel angle (estimated absolute angle)	—	STR_AB_EST
STR_ROT_SPD	°/s	Steering wheel rotation speed	—	STR_ROT_SPD
STR_TRQ_S_M	Nm	Steering shaft torque (Main)	—	STR_TRQ_S_M
STR_TRQ_S_S	Nm	Steering shaft torque (Sub)	—	STR_TRQ_S_S

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
TOTAL_DIST	km, ft, mi	Accumulated total traveled distance from completion of vehicle until EPS control module detects DTC (Odometer value in instrument cluster)	The distance traveled when the EPS control module detected a DTC can be calculated by performing the following procedure. 1. Verify the odometer value in the instrument cluster. 2. Verify the snap shot data item TOTAL_DIST. 3. Subtract 2 from 1.	—
TOTAL_TIME	hh:mm:ss*1	Accumulated total elapsed time since vehicle completion until EPS control module detects a DTC **Note** • When the ROOM fuse is removed, and the ignition is switched off, the time is not included in the elapsed time.	The elapsed time when the EPS control module detected a DTC can be calculated by performing the following procedure. 1. Verify the PID item TOTAL_TIME of the instrument cluster. 2. Verify the snap shot data item TOTAL_TIME. 3. Subtract 2 from 1.	—
VPWR	V	Power supply	—	VPWR
VSPD	KPH, MPH	Vehicle speed	—	VSPD
VSPD_STATUS	Stop/ 0 - 10 km/h/ Over 10 km/h/ FAIL	Vehicle speed status	• The EPS control module constantly receives the vehicle speed sent via CAN signal from the instrument cluster. • If a DTC is detected, the EPS control module records the vehicle speed when the DTC was detected, and it is displayed in the M-MDS.	—

*1 : Fractions of seconds may be indicated after the decimal point

PID/data monitor function

- The PID/data monitor function is used for optionally selecting input/output signal monitor items preset in the EPS control module and reading them out in real-time.

—: Not applicable

Monitor item (Mazda Modular Diagnostic System (M-MDS) display)	Definition	Unit/operation
ECU_IN_TMP	ECU internal temperature	°C, °F
ENG_RPM	Engine speed	RPM
MT_CURRENT	EPS motor control current	A
OH_CR_C	Current overheat protection control (complete)	Yes/No
OH_CR_M	Current overheat protection control (middle)	Yes/No
OH_HIST_C	History of overheat protection control (complete)	Yes/No
OH_HIST_M	History of overheat protection control (middle)	Yes/No

Monitor item (Mazda Modular Diagnostic System (M-MDS) display)	Definition	Unit/operation
OH_IG_CNT_C	The number of times the ignition is switched ON during the period from when overheat protection control (complete) is finished to the time the next overheat protection control (complete) is started • Displays a maximum number of 255 times in which the ignition is switched ON until the first overheat protection control (complete) is started. • During overheat protection control (complete), the number of times the ignition is switched ON is not counted, and 0 is displayed.	—
OH_IG_CNT_M	The number of times the ignition is switched ON during the period from when overheat protection control (middle) is finished to the time the next overheat protection control (middle) is started • Displays a maximum number of 255 times in which the ignition is switched ON until the first overheat protection control (middle) is started. • During overheat protection control (middle), the number of times the ignition is switched ON is not counted, and 0 is displayed.	—
STR_AB_EST	Steering wheel angle (estimated absolute angle)	°
STR_ANG	Steering wheel angle	°
STR_ROT_SPD	Steering wheel rotation speed	°/s
STR_TRQ_S_M	Steering shaft torque (main)	Nm
STR_TRQ_S_S	Steering shaft torque (sub)	Nm
VPWR	Power supply	V
VSPD	Vehicle speed	KPH, MPH

External tester communication function

- The external tester communication function enables communication of diagnostic data (DTC read-outs, input/output signal read-outs) between the EPS control module and an external tester.
- By connecting the Mazda modular diagnostic system (M-MDS) to the DLC-2, diagnostic data can be sent and received between the Mazda modular diagnostic system (M-MDS) and the EPS control module via CAN lines.
- The EPS control module receives the command signals of the malfunction detection function and PID/data monitor function from the Mazda modular diagnostic system (M-MDS), and sends DTCs and data regarding the operating status of each input/output part to the Mazda modular diagnostic system (M-MDS).

Diagnostic function name	Signal received	Signal sent
Malfunction detection function	DTC verification signal	DTC(s)
PID/data monitor function	Command signal to read selected monitor item	Monitored data for requested monitor item